LOCAL FINITE TYPENESS DESCENDS FROM A COMPOSITE MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is short and consists of the diagram plus the two named inputs; the author presents it as 'at this point we can use ... to conclude'.

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Lifting a chain of morphisms of algebraic spaces to a chain of schemes. Spaces, Lemma spaces-lemma-lift-morphism-presentations, used at proof lines 4714-4723. Produces the three-column diagram.
- The local characterization of local finite typeness for algebraic spaces. Lemma lemma-finite-type-local (4614-4653), used at proof lines 4724-4727. Allows the statement to be checked on the presentation.
- The scheme case: local finite typeness descends from a composite. Morphisms, Lemma morphisms-lemma-permanence-finite-type, used at proof lines 4726-4728. Is the input applied to U -> V -> W.

Source reference. Original lemma 040Y: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5273–5273. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 01T8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2365–2365. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02X1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3824–3824. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$, $g : Y \to Z$ be morphisms of algebraic spaces over S. If $g \circ f : X \to Z$ is locally of finite type, then $f : X \to Y$ is locally of finite type.*

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. We can find a diagram

$$\begin{array}{ccccc} U & \longrightarrow & V & \longrightarrow & W \\ \downarrow & & \downarrow & & \downarrow \\ X & \longrightarrow & Y & \longrightarrow & Z \end{array}$$

where U, V, W are schemes, the vertical arrows are étale and surjective, see Spaces, Lemma 02X1. At this point we can use Lemma 040Y and Morphisms, Lemma 01T8 to conclude. □